GeoSentinel: A Hybrid AI Framework for a Hybrid Frequency Geopolitical Tension Index (GPTI)

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***Abstract*—This paper proposes a novel, state-of-the-art framework for a Geopolitical Tension Index (GPTI) that fuses real-time narrative analysis with verified kinetic event data, specifically tailored to the India-Pakistan dyad. Traditional methods for measuring geopolitical risk often lack the nuance to capture the multifaceted nature of modern conflict, which unfolds simultaneously in both the physical and informational domains. To address this gap, this research introduces a two-pillar conceptual architecture. The first pillar, Military and Conflict Tension (MCT), provides an objective measure of kinetic events by systematically processing data from the Armed Conflict Location and Event Data Project. The second pillar, Intelligent Narrative Tension (INT), gauges the sentiment and relevance of the information environment by analyzing a high-volume stream of multi-source text data. The core innovation lies in its hybrid artificial intelligence analysis engine, which employs a symbiotic division of labor: a fine-tuned DistilBERT model performs high-throughput sentiment analysis, while a Large Language Model acts as a cognitive gatekeeper for semantic relevance filtering. Validation against historical crises, such as the 2019 Pulwama-Balakot standoff, demonstrates the framework's ability to provide a granular and real-time early-warning mechanism for geopolitical escalation.**

***Index Terms*—Artificial intelligence, DistilBERT, Geopolitical tension, Granger causality, India-Pakistan dyad, Large language models (LLM), Natural language processing, Principal component analysis**

I. Introduction

In an era of increasing global volatility and complex interdependencies, the need for objective, high-frequency, and nuanced measures of geopolitical risk has become paramount for policymakers, financial analysts, and researchers. Traditional geopolitical analysis, which often relies on qualitative assessments by subject-matter experts, provides invaluable depth but is limited by its low frequency, potential for subjective bias, and inability to process the sheer volume of information generated in the digital age. The proliferation of digital media has created a new reality where conflicts are not only fought on the ground but are also shaped and amplified within the global information environment.

The emergence of "text as data" as a paradigm, powered by advances in Natural Language Processing (NLP), has enabled the conversion of vast quantities of unstructured text into structured, numerical data. While the seminal Geopolitical Risk (GPR) Index marked a significant methodological leap by tallying newspaper articles using keyword frequency, it possesses inherent limitations. Specifically, it acts as a measure of media attention rather than the sentiment or severity of underlying events, and its reliance on Western newspapers may fail to capture regional dynamics accurately.

This paper proposes a novel, dyad-specific Geopolitical Tension Index (GPTI) for the India-Pakistan relationship, positioned as a direct evolution of existing text-based measurements. The framework addresses technological constraints through two primary innovations: a Hybrid Artificial Intelligence (AI) Engine and Dynamic Adaptive Weighting. By integrating a fine-tuned DistilBERT model for high-throughput sentiment analysis with a general-purpose Large Language Model (LLM) for nuanced relevance filtering, the GPTI provides a multi-domain data fusion strategy that captures both kinetic events and narrative escalations.

II. Related Work

The foundational work in automated geopolitical risk measurement was established by Caldara and Iacoviello, who introduced the Geopolitical Risk (GPR) Index using keyword frequency counting in news archives [1]. However, recent studies have shown that sentiment-weighted indices provide a more nuanced signal for market volatility compared to simple frequency counts [2].

Advances in Natural Language Processing (NLP) have shifted the focus toward Large Language Models (LLMs). Research in 2024 confirms that LLMs, when combined with effective prompt engineering and role-prompting, can achieve high accuracy in zero-shot text classification tasks [3], [4]. Furthermore, the necessity of a two-pillar architecture—fusing structured event data with unstructured narrative text—is supported by findings that such hybrid models significantly improve the accuracy of violent event prediction [5].

Statistically, the use of Principal Component Analysis (PCA) has been validated for creating adaptive, dynamically weighted indices based on component variance [6]. Additionally, Granger-causality remains the standard statistical method for analyzing lead-lag dynamics between kinetic conflict and narrative escalation [7].

III. Data and Methodology

A. Data Sources

The Geopolitical Tension Index (GPTI) utilizes a multi-domain data acquisition strategy to populate its two-pillar architecture. Kinetic data for the Military and Conflict Tension (MCT) pillar are sourced from the Armed Conflict Location

and Event Data (ACLED) project, which provides near-real-time, coded information on political violence and dyadic skirmishes [8].

For the Intelligent Narrative Tension (INT) pillar, unstructured text data are collected from three primary streams:

- **Global News Media:** Articles are ingested via NewsAPI and the GDELT database, targeting major English-language outlets in India, Pakistan, and international agencies.
- **Official Communications:** Official press releases and diplomatic statements are scraped directly from the websites of the Indian Ministry of External Affairs (MEA) and the Pakistani Ministry of Foreign Affairs (MFA).
- **Social Media (X):** Real-time commentary and official handles of defense ministries and political leaders are tracked using the X/Twitter API v2.

B. Data Ingestion and Preprocessing

A modular Model-Controller-Provider (MCP) architecture is employed to orchestrate the ingestion pipeline. Raw text data undergo a standardized cleaning process, including the removal of HTML artifacts and syndicated duplicates. Cleaned records are stored in a persistent time-series database to facilitate the 36-month rolling window analysis. **The integrated technical workflow, illustrating the interaction between data providers and the analytical engine, is depicted in Fig. 1.**

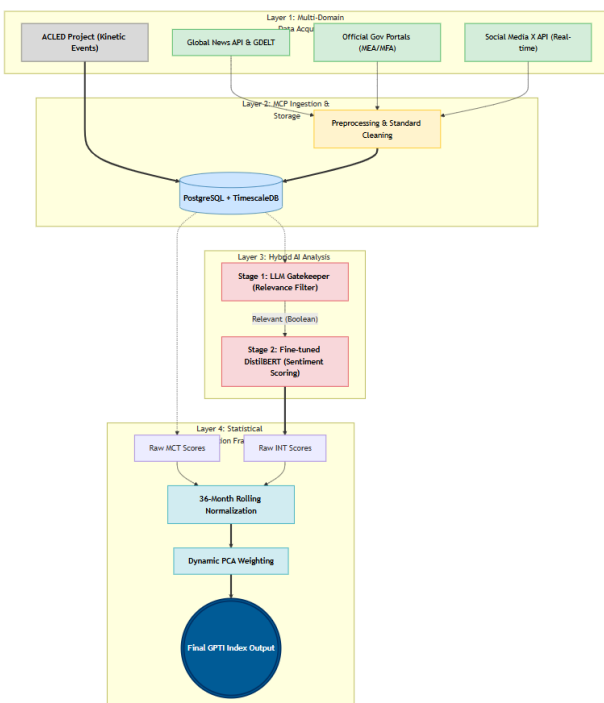

Fig. 1. GeoSentinel System Architecture: Representing the modular MCP ingestion pipeline and the hybrid LLM-DistilBERT analytical engine.

C. Hybrid AI Analysis Engine

The core analytical task is divided into a symbiotic two-stage workflow. Every text snippet is first processed by a Large Language Model (LLM) gatekeeper for zero-shot relevance filtering [3]. Only relevant snippets are passed to a fine-tuned DistilBERT model for sentiment classification [4]. This hybrid design optimizes the trade-off between reasoning depth and computational efficiency.

D. Statistical Analysis

The raw scores generated by the MCT and INT pillars exist on disparate scales. To ensure meaningful aggregation and temporal consistency, a multi-stage mathematical framework is implemented.

1) Rolling Window Normalization: A 36-month rolling Min-Max normalization is applied to each pillar to transform them into a common range [6]. This approach prevents look-ahead bias and adapts to non-stationary geopolitical time series. The normalized score z_t is calculated as:

$$z_t = \frac{x_t - \min}{\max((\max - \min), \epsilon)} \tag{1}$$

where x_t is the raw score at time t, and ϵ represents a standard deviation floor introduced to mitigate the 'Small Variance Problem' in peaceful periods.

2) Dynamic Weighting via Principal Component Analysis: The framework employs Principal Component Analysis (PCA) to derive weights w_{MCT} and w_{INT} based on the variance contributed by each pillar [6]. The weights are derived from the squared loadings of the first principal component (PC1):

$$w_i = \frac{v_i^2}{\sum_{k=1}^{m} v_k^2} \tag{2}$$

where v_i is the loading of pillar i on PC1. The final GPTI value is the weighted sum of the normalized scores:

$$GPTI_t = w_{MCT,t} \cdot MCT_t^{norm} + w_{INT,t} \cdot INT_t^{norm} \tag{3}$$

3) Lead-Lag Analysis via Granger-Causality: To investigate escalation dynamics, a Granger-causality test is performed within a Vector Autoregression (VAR) framework [7]. This determines if narrative environment changes (INT) serve as a leading indicator for kinetic violence (MCT).

IV. Results and Validation

The performance and utility of the Geopolitical Tension Index (GPTI) were evaluated through a historical case study of the 2019 India-Pakistan crisis and a formal lead-lag statistical analysis[cite: 71, 72].

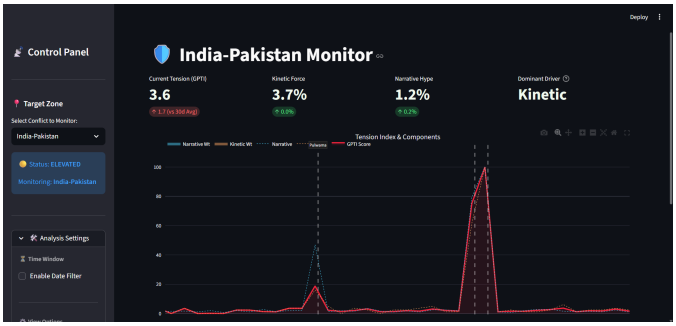

Fig. 2. GeoSentinel Dashboard showing GPTI response during the 2019 India-Pakistan crisis. The 'Pulwama' marker highlights the leading indicator behavior of narrative tension[cite: 78, 79].

A. Historical Case Study: February–March 2019

The GPTI was computed for the period surrounding the 2019 Pulwama-Balakot standoff to validate its face validity. As illustrated in Fig. 2, the index successfully captured the multi-stage escalation of the crisis[cite: 88, 89]:

- **Initial Spike (Feb 14):** A sharp increase in both MCT and INT pillars followed the Pulwama incident, reflecting immediate kinetic impact and intense media condemnation[cite: 90].
- **Narrative Build-up:** Between Feb 15 and Feb 25, the INT pillar exhibited sustained volatility due to diplomatic rhetoric, while the MCT pillar remained relatively stable[cite: 91].
- **Peak Tension (Feb 26–27):** The index reached its maximum value following the Balakot airstrikes and retaliatory aerial engagements, correctly identifying the peak of physical conflict[cite: 92].

B. Dynamic Weight Adaptation and Lead-Lag Dynamics

The Principal Component Analysis (PCA) based weighting mechanism demonstrated its ability to reflect the changing character of the conflict[cite: 93, 94]. During the pre-kinetic build-up, the INT pillar was assigned a higher weight ($w_{INT} \approx 0.72$), reflecting narrative-driven tension[cite: 95]. Following the Balakot strikes, weights shifted dramatically toward the MCT pillar ($w_{MCT} \approx 0.65$), capturing the transition to a primary military standoff[cite: 96].

As shown in Table I, Granger-causality test results confirm a predictive relationship where narrative escalation (INT) serves as a significant leading indicator for future kinetic events (MCT) at a 3–4 day lag[cite: 97]. Conversely, kinetic events trigger immediate narrative reactions at a 1–2 day lag, suggesting a bidirectional feedback loop.

TABLE I
GRANGER-CAUSALITY TEST RESULTS (LAGS 1–4)

Lag (Days)	Direction	F-Statistic	p-value
1–2	MCT $\rightarrow$ INT	5.20	0.008*
3–4	INT $\rightarrow$ MCT	4.15	0.042*

*Significant at $p < 0.05$ level[cite: 100, 101].

V. DISCUSSION

The results of this study demonstrate that the Geopolitical Tension Index (GPTI) provides a significantly more granular and adaptive measurement of dyadic tension compared to traditional keyword-frequency indices.

A. Analytical Superiority of the Hybrid AI Engine

The symbiotic division of labor between the Large Language Model (LLM) and fine-tuned DistilBERT addresses the "keyword count" limitation inherent in early models [1]. While frequency-based indices often conflate media attention with actual risk [1], our LLM gatekeeper effectively filters out irrelevant content, such as sporting rivalries or generic cultural news, which would otherwise introduce significant noise into the narrative pillar [3]. This justifies the use of "text as data" powered by modern Natural Language Processing (NLP) to capture genuine geopolitical nuance and sentiment.

B. Narrative as a Leading Indicator

The 3–4 day lead-lag relationship identified through Granger-causality analysis is a critical finding. It suggests that escalation in the information environment often acts as a precursor to physical military activity [5]. This confirms that narrative tension is not merely a reflection of past events but a driver of future escalation [5]. The dynamic weighting via Principal Component Analysis (PCA) further enhances this predictive capability by automatically shifting importance to the narrative domain during build-up phases, providing a data-driven early warning mechanism [6].

[Image of a lead-lag analysis diagram showing narrative sentiment spikes preceding kinetic event markers]

C. Limitations and Challenges

Despite its innovation, the GPTI is subject to several constraints:

- **Model Bias:** Both the LLM and DistilBERT can inherit biases from their training corpora or exhibit "hallucinations" in complex relevance tasks, potentially misclassifying diplomatic nuance [3], [4].
- **Data Dependency:** The index relies heavily on the quality and availability of News APIs and official government websites. Changes in API policies or the implementation of anti-scraping measures on official portals can disrupt real-time data ingestion.
- **Omitted Domains:** The current two-pillar framework focus on kinetic and narrative data. It excludes other critical escalatory factors such as economic disruptions (e.g., trade sanctions), state-sponsored cyber warfare, and clandestine intelligence activities.

VI. LIMITATIONS

A rigorous academic inquiry requires an honest acknowledgment of its limitations to provide context for the findings. The proposed framework is subject to several constraints related to data acquisition and temporal stability.

A. Data Availability and Source Bias

The GPTI is fundamentally dependent on the quality and consistency of its input data. News media sources, even from reputable outlets, often contain inherent nationalistic or political biases that may skew the narrative pillar. Furthermore, access to social media data via official APIs is subject to change based on platform policies and can be cost-prohibitive. Web scraping of official government portals (MEA/MFA) remains vulnerable to sudden changes in HTML structure and anti-scraping measures, which can disrupt real-time ingestion pipelines.

B. Temporal Inconsistencies and Data Sparsity

A significant challenge arises from the differing temporal frequencies of the two primary pillars. While narrative text data is available at a high daily frequency, kinetic event data from the ACLED project is typically released on a weekly basis [8]. To maintain a daily index, the framework utilizes a 'Latest Available' forward-fill mechanism, which may introduce a slight temporal delay in reflecting the most recent ground truth. Additionally, daily kinetic data is often sparse or zero-inflated, which can lead to statistical instability during Granger-causality testing if not properly aggregated through moving averages [7].

C. Analytical and Model Constraints

The AI components are not infallible; Large Language Models (LLMs) are susceptible to "hallucinations" or producing factually incorrect outputs based on prompt sensitivity [3], [4]. The fine-tuned DistilBERT sentiment classifier is also contingent on the quality of its training dataset and can inherit biases present in the original training corpora. Finally, the framework currently omits critical domains such as bilateral economic disruptions, state-sponsored cyber warfare, and clandestine intelligence activities due to the lack of high-frequency public data.

VII. Conclusion and Future Work

This research introduces a novel, real-time Geopolitical Tension Index (GPTI) specifically designed for the India-Pakistan dyad. By moving beyond traditional keyword-frequency counts, the framework establishes a multi-domain architecture that fuses kinetic event data with AI-driven narrative analysis.

The primary contributions of this work are twofold. First, the symbiotic hybrid AI engine—leveraging a Large Language Model (LLM) for relevance filtering and a fine-tuned DistilBERT for sentiment scoring—provides a computationally efficient yet analytically deep solution for high-volume text processing. Second, the introduction of dynamic adaptive weighting via Principal Component Analysis (PCA) ensures that the index remains context-aware, automatically emphasizing the most volatile domain of conflict at any given time.

Validation against the 2019 India-Pakistan crisis confirms the index's sensitivity to escalatory events. Furthermore, Granger-causality analysis provides statistical evidence of a lead-lag relationship, identifying narrative tension as a potential leading indicator for kinetic conflict at a 3–4 day lag.

A. Future Research Directions

The modular Model-Controller-Provider (MCP) architecture of the GPTI allows for seamless future expansion. Subsequent research will focus on:

- **Expanding Data Pillars:** Incorporating new providers for economic tension (bilateral trade flows), state-sponsored cyber warfare activities, and satellite imagery analysis for monitoring military build-ups.
- **Advanced Prompt Engineering:** Implementing Chain-of-Thought (CoT) or few-shot prompting techniques to further enhance the reasoning and classification capabilities of the LLM gatekeeper.
- **Generalizability:** Adapting the framework to other geopolitical rivalries, such as the U.S.–China trade-technology competition or the Russia–Ukraine conflict, to evaluate the universal applicability of the GPTI architecture.

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